

The Fundamental Theorem of Calculus

A Graduate Introduction

Part 1: The Connection

If f is continuous on $[a, b]$ and $F(x) = \int_a^x f(t) dt$, then

$$F'(x) = f(x)$$

This means **integration and differentiation are inverse operations.**

Summary

- FTC Part 1: the derivative of an integral recovers the integrand
- FTC Part 2: $\int_a^b f(x) dx = F(b) - F(a)$
- These results hold for any continuous f